Matrix Lie groups

$G \leq GL(n, \mathbb{C})$ closed under the topology of $GL(n,\mathbb{C})$

↳ subgroup

trick: if $f: X \to Y$ continuous, the pre-image of a closed set is closed

$SL(n,\mathbb{C}) = \{ A \in GL(n,\mathbb{C}) \mid \det(A) = 1 \}$

$\det : M_n(\mathbb{C}) \to \mathbb{C}$ continuous

topology on $GL(n,\mathbb{C})$ is the one induced from $M_n(\mathbb{C})$

$M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$.

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ex: $GL(n,\mathbb{C})$ general linear group (complex)

$SL(n,\mathbb{C})$ special linear group (complex)

$GL(n,\mathbb{R})$ general linear group (real)

$= \{ A \in GL(n,\mathbb{C}) \mid \mathrm{Im}(A) = 0 \}$

$= \mathrm{Im}^{-1}(\{0\})$

imaginary part

$\mathrm{Im} : A \in M_n(\mathbb{C}) \mapsto \frac{A - \bar{A}}{2i} \in M_n(n,\mathbb{R})$

real $\Leftrightarrow \mathrm{Im}(A) = 0$

$SL(n,\mathbb{R})$ special linear group (real)

$O(n) = \{ A \in GL(n,\mathbb{R}) \mid A^T A = \mathbb{1} \}$

→ orthogonal group

$A \to A^T A$ continuous

pre-image of $\{\mathbb{1}\}$

$SO(n) = \{ A \in O(n) \mid \det(A) = 1 \}$ special orthogonal group

$U(n) = \{ A \in GL(n,\mathbb{C}) \mid A^* A = \mathbb{1} \}$

or $A^\dagger = \bar{A}^T$ (conjugate transpose or adjoint)

→ unitary group

$SU(n)$ special unitary group

Associated Lie algebras

→ idea: want to deal with infinitesimal group elements

$g \in GL(n, \mathbb{C})$ $\qquad g = \mathbb{1} + \varepsilon X + O(\varepsilon^2)$ → lives in Lie algebras

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<u>Matrix exponential</u> $\qquad$ if $X \in M_n(\mathbb{C})$ then $e^X = \sum_{k=0}^{\infty} \frac{1}{k!} X^k$

- $\det(e^X) = e^{\operatorname{tr} X} \neq 0 \Rightarrow e^X$ invertible, $e^X \in GL(n, \mathbb{C})$
- $(e^X)^{-1} = e^{-X}$
- $e^{X+Y} \neq e^X e^Y$ unless X, Y commute

every element of $GL(n, \mathbb{C})$ is of the form e^X for $X \in M_n(\mathbb{C})$

if $g_\varepsilon = e^{\varepsilon X}$ $\quad \varepsilon \in \mathbb{R}$ $\quad$ (one-parameter <u>subgroup</u>)

$\{ g_\varepsilon \mid \varepsilon \in \mathbb{R} \} \leq GL(n, \mathbb{C})$ $\qquad e^{\varepsilon_1 X} e^{\varepsilon_2 X} = e^{(\varepsilon_1 + \varepsilon_2) X}$

$$g_\varepsilon = g_0 + \varepsilon \frac{d}{d\varepsilon}(e^{\varepsilon X})\Big|_{\varepsilon=0} + O(\varepsilon^2)$$

$$= \mathbb{1} + \varepsilon X + O(\varepsilon^2)$$

→ Lie algebra elements

$$\frac{d}{d\varepsilon} e^{\varepsilon X} = \underbrace{e^{\varepsilon X}}_{\to 1} X$$

$G \leq GL(n, \mathbb{C})$ $\quad$ MLG

define $\quad \mathrm{Lie}(G) = \{ X \in M_n(\mathbb{C}) \mid e^{\varepsilon X} \in G \ \forall \varepsilon \in \mathbb{R} \}$

$X \in \mathrm{Lie}(G) \Leftrightarrow X$ generates a 1-parameter subgroup of G

why is it an algebra?

- vector space
- has an operation (compatible with v. space structure)

$\mathrm{Lie}(G)$ is a real vector space

and the operation is the Lie bracket

$$[\cdot,\cdot] : \mathrm{Lie}(G) \times \mathrm{Lie}(G) \to \mathrm{Lie}(G) \qquad \text{(commutator)}$$
$$(X, Y) \qquad XY - YX$$

$[\cdot,\cdot]$ bilinear

operation compatible if $[\alpha X + \beta Y, Z] = \alpha [X, Z] + \beta [Y, Z]$

if $X \in \mathrm{Lie}(G)$ then $\alpha X \in \mathrm{Lie}(G)$ $\alpha \in \mathbb{R}$

if $X, Y \in \mathrm{Lie}(G)$ then $X + Y \in \mathrm{Lie}(G)$

<u>Examples</u> Notation $\mathrm{Lie}(G) = \mathfrak{g}$ or $\underline{g}$

$$\underline{gl}(n, \mathbb{C}) = \{ X \in M_n(\mathbb{C}) \mid e^{\varepsilon X} \in GL(n, \mathbb{C}) \ \forall \varepsilon \in \mathbb{R} \}$$
$$= M_n(\mathbb{C}) \quad \text{vector space!}$$

$e^{\varepsilon X}$ always invertible
$\to$ belongs to $GL(n, \mathbb{C})$

$$\underline{sl}(n, \mathbb{C}) = \{ X \in M_n(\mathbb{C}) \mid e^{\varepsilon X} \in SL(n, \mathbb{C}) \ \forall \varepsilon \in \mathbb{R} \}$$

$$1 = \det(e^{\varepsilon X}) = e^{\mathrm{tr}(\varepsilon X)} = e^{\varepsilon \, \mathrm{tr} X} \qquad 1 = e^{\varepsilon \, \mathrm{tr} X}$$

take derivative w.r.t. ε on both sides, and set $\varepsilon = 0$

$$0 \Big|_{\varepsilon=0} = \mathrm{tr} X \, e^{\varepsilon \, \mathrm{tr} X} \Big|_{\varepsilon=0} \Rightarrow \mathrm{tr}(X) = 0$$

<u>Proposition:</u> $X, Y \in M_n(\mathbb{C})$

$$e^{\varepsilon X} = e^{\varepsilon Y} \quad \forall \varepsilon \in \mathbb{R} \iff X = Y$$

<u>proof</u>

$\Leftarrow$ if $X = Y$ then $e^{\varepsilon X} = e^{\varepsilon Y}$ $\forall \varepsilon$

$\Rightarrow$ if $e^{\varepsilon X} = e^{\varepsilon Y}$ $\forall \varepsilon \in \mathbb{R}$ then

$$\underbrace{\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} e^{\varepsilon X}}_{= X} = \underbrace{\frac{d}{d\varepsilon}\Big|_{\varepsilon=0} e^{\varepsilon Y}}_{= Y} \Rightarrow X = Y$$

$$1 = e^{\varepsilon \operatorname{tr} X} = e^{0} = e^{\varepsilon \operatorname{tr} X} \Rightarrow \operatorname{tr} X = 0$$

$\underline{\mathfrak{sl}}(n, \mathbb{C}) = \{ X \in M_n(\mathbb{C}) \mid \operatorname{tr} X = 0 \}$ vector space

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$\mathfrak{gl}(n, \mathbb{R}) = \{ X \in M_n(\mathbb{C}) \mid e^{\varepsilon X} \in GL(n, \mathbb{R}) \ \forall \varepsilon \}$

<u>note:</u> $\overline{e^{\varepsilon X}} = e^{\overline{\varepsilon X}} = e^{\varepsilon \overline{X}}$

$$e^{\varepsilon X} \in GL(n, \mathbb{R}) \ \forall \varepsilon \iff e^{\varepsilon X} = \overline{e^{\varepsilon X}} \ \forall \varepsilon$$

$$\iff e^{\varepsilon X} = e^{\varepsilon \overline{X}} \ \forall \varepsilon$$

$\underline{\mathfrak{gl}}(n, \mathbb{R}) = M_n(\mathbb{R})$ vector space

$\underline{\mathfrak{sl}}(n, \mathbb{R}) = \{ X \in M_n(\mathbb{R}) \mid \operatorname{tr}(X) = 0 \}$

$\underline{\mathfrak{o}}(n) = \{ X \in M_n(\mathbb{R}) \mid e^{\varepsilon X} \in O(n) \ \forall \varepsilon \in \mathbb{R} \}$

$$e^{\varepsilon X} \in O(n) \ \forall \varepsilon \in \mathbb{R} \iff (e^{\varepsilon X})^T e^{\varepsilon X} = \mathbb{1} \ \forall \varepsilon$$

$$\iff e^{\varepsilon X^T} e^{\varepsilon X} = \mathbb{1} \ \forall \varepsilon$$

$$\iff e^{\varepsilon X^T} = e^{-\varepsilon X} \ \forall \varepsilon$$

$\underline{\mathfrak{o}}(n) = \{ X \in M_n(\mathbb{R}) \mid X^T = -X \}$ vector space

if $X^T = -X$ then $\operatorname{tr}(X) = 0$

$\underline{\mathfrak{so}}(n) = \underline{\mathfrak{o}}(n)$